IMAT 모의고사 3회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Sleep researchers once treated dreaming as a by-product of random brain activity. More recent work suggests that sleep, particularly slow-wave sleep, plays an active role in consolidating memories: brain regions active during learning are reactivated during subsequent sleep. In experiments, volunteers who slept after learning a task outperformed those who stayed awake for the same period. However, whether dreams themselves contribute to this process, or merely accompany it, remains disputed.

1. Which of the following is presented in the passage as still uncertain?

A. Whether volunteers who slept performed better.
B. Whether slow-wave sleep exists.
C. Whether brain regions are reactivated during sleep.
D. Whether sleep helps memory consolidation.
E. Whether dreams play a causal role in memory consolidation.

2. The experiment described compares volunteers who slept with volunteers who stayed awake "for the same period". Why is this detail important?

A. It shows that waking impairs the brain.
B. It controls for the time elapsed between learning and testing.
C. It proves dreams are random.
D. It ensures that both groups dreamed.
E. It ensures that both groups learned different tasks.

3. Which disease was declared eradicated worldwide by the World Health Organization in 1980?

A. Measles
B. Smallpox
C. Malaria
D. Polio
E. Tuberculosis

4. Who wrote *The Divine Comedy*?

A. Ariosto
B. Machiavelli
C. Dante Alighieri
D. Petrarch
E. Boccaccio

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "Only students who pass the exam can enrol in the course. Marta enrolled in the course." What can be concluded?

A. All students who passed enrolled.
B. Marta did not take the exam.
C. Nothing can be concluded.
D. Marta failed the exam.
E. Marta passed the exam.

6. A solution is diluted so that its concentration falls by half every hour. After 4 hours, what fraction of the original concentration remains?

A. 1/16
B. 1/8
C. 1/12
D. 1/32
E. 1/4

7. Anna is older than Bruno. Carla is younger than Bruno but older than Dario. Elena is older than Anna. Who is the second youngest?

A. Anna
B. Carla
C. Dario
D. Elena
E. Bruno

8. A box has a volume of 240 cm^3. Its length is 10 cm and its width is 6 cm. What is its height?

A. 4 cm
B. 6 cm
C. 2 cm
D. 3 cm
E. 24 cm

9. In a survey, 60% of respondents were women, and 25% of the women smoke. If 300 people were surveyed, how many women smoke?

A. 180
B. 60
C. 45
D. 30
E. 75

Section 3 — Biology (Q10–32)

10. Which statement about the Golgi apparatus is correct?

A. It is found only in plant cells.
B. It produces ATP.
C. It modifies, sorts and packages proteins received from the endoplasmic reticulum.
D. It contains the cell's DNA.
E. It synthesises lipids for the nucleus.

11. In eukaryotic mRNA processing, which of these does NOT normally occur?

A. Export from the nucleus
B. Splicing out of introns
C. Addition of a 5' cap
D. Removal of exons
E. Addition of a poly-A tail

12. Which of the following is a correct comparison of mitosis and meiosis in humans?

A. Both produce four daughter cells.
B. Mitosis produces haploid cells; meiosis produces diploid cells.
C. Mitosis produces genetically identical cells; meiosis produces genetically varied haploid cells.
D. Both involve two nuclear divisions.
E. Crossing over is typical of mitosis.

13. In which location does the Krebs cycle occur in eukaryotic cells?

A. Intermembrane space
B. Mitochondrial matrix
C. Inner mitochondrial membrane
D. Cytoplasm
E. Nucleus

14. Haemophilia A is X-linked recessive. An affected man has children with a homozygous normal woman. Which is true?

A. Half of the daughters will be affected.
B. All sons will be affected.
C. All children will be unaffected and non-carriers.
D. All daughters will be carriers.
E. Half of the sons will be carriers.

15. Which of the following is a product of the light-dependent reactions of photosynthesis?

A. Pyruvate
B. Ribulose bisphosphate
C. CO_2
D. Glucose
E. ATP and NADPH

16. The main function of red blood cells is to:
A. produce antibodies.
B. digest pathogens.
C. clot blood.
D. fight infection.
E. transport oxygen using haemoglobin.

17. Which of the following molecules is NOT a lipid?
A. Testosterone
B. Phospholipid
C. Cholesterol
D. Glycogen
E. Triglyceride

18. The myelin sheath in the peripheral nervous system is formed by:
A. Oligodendrocytes
B. Microglia
C. Neurons themselves
D. Schwann cells
E. Astrocytes

19. Which valve prevents backflow of blood from the left ventricle into the left atrium?
A. Tricuspid valve
B. Aortic valve
C. Mitral (bicuspid) valve
D. Pulmonary valve
E. Semilunar valve of the vena cava

20. In snapdragons, flower colour shows incomplete dominance: RR is red, rr is white, Rr is pink. Crossing two pink plants gives which phenotypic ratio?
A. 1 red : 2 pink : 1 white
B. 3 red : 1 white
C. All pink
D. 9 : 3 : 3 : 1
E. 1 red : 1 white

21. Thyroxine production depends on adequate dietary intake of:
A. Calcium
B. Iron
C. Zinc
D. Sodium
E. Iodine

22. The DNA of a species contains 30% adenine. What percentage is guanine?
A. 60%
B. 20%
C. 30%
D. 70%
E. 40%

23. Which cells directly kill virus-infected body cells?

A. Mast cells
B. Erythrocytes
C. Cytotoxic ($CD8^+$) T cells
D. Helper T cells
E. Plasma cells

24. In the loop of Henle, which part is permeable to water but largely impermeable to solutes?

A. Descending limb
B. Glomerulus
C. Thin ascending limb
D. Macula densa
E. Thick ascending limb

25. A diploid organism has 2n = 8. How many different combinations of chromosomes can be produced in gametes by independent assortment alone?

A. 4
B. 256
C. 8
D. 16
E. 64

26. Which bond links amino acids in a polypeptide?

A. Phosphodiester
B. Peptide
C. Ester
D. Hydrogen
E. Glycosidic

27. Which of the following is an example of a chromosomal aneuploidy?

A. Huntington's disease
B. Sickle cell anaemia
C. Cystic fibrosis
D. Down syndrome (trisomy 21)
E. Haemophilia

28. A red blood cell is placed in a hypotonic solution. What happens?

A. It actively pumps water out and stays the same.
B. It divides.
C. Nothing happens.
D. It shrinks (crenates).
E. It swells and may burst (haemolysis).

29. Which structure is used by many bacteria for movement?

A. Nucleoid
B. Flagellum
C. Plasmid
D. Pili
E. Capsule

30. Pepsin is an enzyme that digests:
A. carbohydrates in the mouth.
B. nucleic acids in the colon.
C. starch in the stomach.
D. proteins in the stomach.
E. fats in the small intestine.

31. During translation, the ribosome moves along the mRNA in which direction?
A. 5' → 3'
B. N-terminus to C-terminus of the mRNA
C. 3' → 5'
D. Both directions alternately
E. It does not move

32. Which of the following best describes homologous structures?
A. Structures produced by convergent evolution.
B. Structures that have lost their function.
C. Structures found only in plants.
D. Structures with common evolutionary origin, possibly different functions.
E. Structures with similar function but different evolutionary origin.

Section 4 — Chemistry (Q33–47)

33. How many molecules are in 0.25 mol of CO_2? ($N_A = 6 \times 10^{23}$ mol^{-1})
A. 2.4×10^{24}
B. 6×10^{23}
C. 1.5×10^{24}
D. 2.4×10^{23}
E. 1.5×10^{23}

34. In the reaction $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$, which species is oxidised?
A. No species is oxidised.
B. I_2
C. Fe^{3+}
D. Fe^{2+}
E. I^-

35. A solution has $[OH^-] = 1 \times 10^{-5}$ mol/L at 25 °C. What is its pH?
A. 9
B. 7
C. 10
D. 14
E. 5

36. Which of the following has the largest ionic radius?
A. Na^+
B. O^{2-}
C. F^-
D. Mg^{2+}
E. Al^{3+}

37. What is the hybridisation of carbon in methane, CH_4?

A. sp^3
B. sp^3d
C. sp^2
D. sp
E. Unhybridised

38. 10 g of $CaCO_3$ (M = 100 g/mol) is heated: $CaCO_3 \rightarrow CaO + CO_2$. What mass of CO_2 (M = 44 g/mol) is produced?

A. 5.6 g
B. 4.4 g
C. 2.2 g
D. 10 g
E. 44 g

39. Which reaction converts an alkene into an alkane?

A. Hydration
B. Hydrogenation
C. Halogenation with Br_2
D. Oxidation
E. Esterification

40. How many grams of glucose ($C_6H_{12}O_6$, M = 180 g/mol) are needed to make 250 mL of a 0.2 mol/L solution?

A. 90 g
B. 0.9 g
C. 9 g
D. 36 g
E. 3.6 g

41. Which substance has a giant covalent structure?

A. Copper
B. NaCl
C. Diamond
D. Iodine
E. Methane

42. Which of the following is a strong acid in water?

A. H_2CO_3
B. HNO_3
C. CH_3COOH
D. NH_4^+
E. HF

43. A catalyst affects a reaction by:

A. providing an alternative pathway with lower activation energy.
B. being permanently consumed.
C. increasing the yield at equilibrium.
D. changing ΔH of the reaction.
E. shifting equilibrium to the right.

44. What is the IUPAC name of $CH_3CH_2CH_2OH$?
A. Ethanol
B. Propanal
C. Propanone
D. Propan-1-ol
E. Propan-2-ol

45. How many electrons can the n = 3 shell hold at most?
A. 2
B. 8
C. 18
D. 32
E. 10

46. Which statement about a galvanic (voltaic) cell is correct?
A. It converts electrical energy into chemical energy.
B. Electrons flow from cathode to anode in the external circuit.
C. Reduction takes place at the cathode.
D. It requires an external power source.
E. The anode is always positive.

47. Which molecule is chiral?
A. CHBrClF
B. CH_2Cl_2
C. CH_3Cl
D. CCl_4
E. CH_3CH_2OH

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is thrown vertically upward at 20 m/s (g = 10 m/s²). What maximum height does it reach?
A. 20 m
B. 200 m
C. 80 m
D. 10 m
E. 40 m

49. A 60 kg person climbs 5 m of stairs in 10 s (g = 10 m/s²). What average power is developed?
A. 30 W
B. 3000 W
C. 600 W
D. 60 W
E. 300 W

50. A 12 V battery is connected across a 4 Ω resistor. What power is dissipated?
A. 3 W
B. 48 W
C. 36 W
D. 12 W
E. 144 W

51. A sound wave travels at 340 m/s and has a wavelength of 0.5 m. What is its frequency?
A. 680 Hz
B. 1360 Hz
C. 0.0015 Hz
D. 340 Hz
E. 170 Hz

52. A gas is heated at constant pressure from 300 K to 600 K. Its volume:
A. quadruples.
B. stays the same.
C. increases by 300 times.
D. halves.
E. doubles.

53. A 2 kg ball moving at 6 m/s collides with and sticks to a stationary 4 kg ball. What is their common speed after the collision?
A. 2 m/s
B. 6 m/s
C. 1 m/s
D. 3 m/s
E. 4 m/s

54. How much heat is needed to raise the temperature of 2 kg of water by 10 °C? (c = 4200 J/(kg·°C))
A. 8400 J
B. 840 J
C. 84 000 J
D. 840 000 J
E. 42 000 J

55. What is $\log_{10}(0.001)$?
A. 2
B. −2
C. 0.001
D. −3
E. 3

56. What is cos(120°)?
A. √3/2
B. 1/2
C. 0
D. −√3/2
E. −1/2

57. A fair coin is tossed 3 times. What is the probability of getting exactly 2 heads?

A. 2/3
B. 1/2
C. 1/4
D. 1/8
E. 3/8

58. What is the distance between the points $(-1, 2)$ and $(2, 6)$?

A. 4
B. 5
C. 7
D. 25
E. 3

59. If $f(x) = 2x^2 - 3x + 1$, what is $f(-1)$?

A. 0
B. 2
C. -4
D. 6
E. 4

60. Two point charges are separated by distance r. If r is doubled, the electrostatic force between them becomes:

A. half as large.
B. 4 times larger.
C. 2 times larger.
D. one quarter as large.
E. unchanged.

정답 및 해설

번호	정답	해설
1	E	꿈이 기억 강화에 기여하는지는 "여전히 논쟁 중".
2	B	학습–시험 사이 경과 시간을 동일하게 통제.
3	B	천연두(1980년 근절 선언).
4	C	단테 알리기에리.
5	E	"오직 합격자만 등록 가능" → 등록했으면 합격.
6	A	$(1/2)^4 = 1/16$.
7	B	나이순: E > A > B > C > D → 두 번째로 어린 사람 Carla.
8	A	$240/(10\times6) = 4$.
9	C	$300\times0.6 = 180$, $\times0.25 = 45$.
10	C	골지체: 단백질 수식·분류·포장.
11	D	제거되는 것은 인트론, 엑손은 남음.
12	C	체세포분열: 동일 유전자 / 감수분열: 다양한 반수체.
13	B	미토콘드리아 기질.
14	D	아버지 X^h → 딸 모두 보인자, 아들은 Y 받아 정상.
15	E	명반응 산물: ATP, NADPH ($+O_2$).
16	E	헤모글로빈으로 O_2 운반.
17	D	글리코겐은 탄수화물.
18	D	말초 = 슈반세포, 중추 = 희소돌기아교세포.
19	C	승모판.
20	A	불완전 우성 1:2:1.
21	E	요오드.
22	B	A=T=30 → G+C=40 → G=20%.
23	C	세포독성 T세포.
24	A	하행지: 물 투과, 용질 거의 불투과.
25	D	$n=4 \rightarrow 2^4 = 16$.
26	B	펩타이드 결합.
27	D	다운 증후군(21번 삼염색체).
28	E	저장액 → 물 유입 → 용혈.
29	B	편모.
30	D	펩신: 위에서 단백질 분해.
31	A	리보솜은 mRNA를 5'→3'으로 읽음.
32	D	상동기관 = 공통 기원, 기능은 다를 수 있음.
33	E	$0.25 \times 6\times10^{23} = 1.5\times10^{23}$.
34	E	$I^-(-1) \rightarrow I_2(0)$: 산화.
35	A	pOH 5 → pH 9.

번호	정답	해설
36	B	등전자(10e⁻) 중 양성자 수 가장 적은 O^{2-}가 가장 큼.
37	A	sp^3.
38	B	0.1 mol → CO_2 0.1 mol = 4.4 g.
39	B	수소화(H_2, Ni/Pt 촉매).
40	C	0.2×0.25 = 0.05 mol × 180 = 9 g.
41	C	다이아몬드(거대 공유결합 구조).
42	B	HNO_3 강산.
43	A	활성화에너지가 낮은 대체 경로.
44	D	프로판-1-올.
45	C	$2n^2 = 18$.
46	C	환원은 cathode. 갈바니 전지에서 전자는 anode→cathode.
47	A	네 치환기가 모두 다른 탄소.
48	A	$h = v^2/2g = 400/20 = 20$ m.
49	E	$mgh/t = 60×10×5/10 = 300$ W.
50	C	$V^2/R = 144/4 = 36$ W.
51	A	$f = v/\lambda = 680$ Hz.
52	E	샤를 법칙 $V \propto T$.
53	A	12 = 6v → v = 2 m/s.
54	C	$Q = mc\Delta T = 2×4200×10 = 84\,000$ J.
55	D	10^{-3}.
56	E	$\cos 120° = -1/2$.
57	E	${}_3C_2/8 = 3/8$.
58	B	$\sqrt{(9+16)} = 5$.
59	D	2+3+1 = 6.
60	D	쿨롱 법칙 $F \propto 1/r^2$.